

Documentation: The Dragon Egg Cave

What it is

A single self-contained HTML file (~13 KB) with all CSS and JavaScript embedded — no external libraries, frameworks, or servers required. It runs entirely in the browser and was published as a Claude Artifact (a hosted webpage) so it can be opened via a link.

Structure

The page is built as four scrollable sections, linked by a sticky top nav:

Section	Purpose
The Cave	Narrative setup — introduces the dragon egg circle and the two "meters" the game will demonstrate
Confusion Game	Interactive entropy demo
Wrongness Game	Interactive cross-entropy demo
Recap	Kid-friendly definitions of entropy and cross-entropy

How the games actually work (the math under the hood)

Confusion Game (Entropy)

- *Uniform mode*: a slider sets the number of eggs (2–16). The code computes entropy as $H = \log_2(n)$, since every egg is equally likely. This value drives both the numeric readout and the width of the "confusion meter" bar.
- *Weighted mode*: four sliders represent "warmth" for four eggs. Their raw values are normalized into probabilities ($p_i = \text{value}_i / \text{sum of all values}$), then true Shannon entropy is calculated: $H = -\sum p(x) \cdot \log_2 p(x)$. As one egg's slider dominates, entropy visibly drops toward 0.

Wrongness Game (Cross-Entropy)

- A "true egg" index is chosen randomly (`Math.random()`) each round and hidden from the player.
- *Human mode*: clicking one egg records a single certain guess. On reveal, if correct the wrongness is 0; if wrong, it's displayed as infinite (mathematically $-\log_2(0)$), since a 100%-confident guess leaves no partial credit.
- *Bot mode*: five sliders let the player spread "belief points" across all eggs. These are normalized to probabilities, and on reveal the code computes $-\log_2(q \text{ at the true egg})$, capped for display once the probability gets very close to zero. This produces a smooth, partial "wrongness" score instead of an all-or-nothing result — the actual reason real models use this instead of simple right/wrong scoring.
- Egg positions in the circle are generated with basic trigonometry ($x = \cos(\text{angle})$, $y = \sin(\text{angle})$) rather than hard-coded, so the layout adapts to any egg count.

Design choices

- **Typography**: "Baloo 2" (rounded, playful) for headings, "Inter" for body text, "JetBrains Mono" for numbers/labels — pulled from Google Fonts.
- **Color palette**: a dark cave theme (deep purples/blacks) with ember-orange and gold accents to match the dragon/fire motif.
- **Responsive layout**: CSS grid/flexbox with breakpoints so it works on both desktop and mobile.

Tech stack summary

Plain HTML + CSS + vanilla JavaScript. No React, no build step, no backend, no data storage — every calculation happens live in the visitor's browser.